

ABHINEET SRIVASTAVA

M.Sc. Physics Student, UPES, Dehradun, India

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RESEARCH INTERESTS

- Black Hole Accretion Physics
- High Energy Astrophysics and Radiative Processes
- Extraterrestrial Planets
- Galaxy Formation, Evolution, and the Role of Dark Matter and Dark Energy
- Computational Astrophysics and Machine Learning Applications in Astronomy

EDUCATION

Master of Science (Physics)	2024 – 2026
UPES, Dehradun, Uttarakhand, India	76% (current)
Bachelor of Science with Honours (Physics)	2021 – 2024
UPES, Dehradun, Uttarakhand, India	72.3%
Class XII (CBSE)	2020 – 2021
B.B.L. Public School, Bareilly, Uttar Pradesh, India	87.6%
Class X (CBSE)	2018 – 2019
B.B.L. Public School, Bareilly, Uttar Pradesh, India	91.8%

ACADEMIC PROJECTS

M.Sc. Thesis UPES, Dehradun, India	Aug'25 – Present
<i>Decoding Disk-Corona-Jet Connections of Stellar Mass Black Holes through Timing and Spectral Analysis</i>	
Supervisor: Dr. Arka Chatterjee (UPES, Dehradun)	
Co-Supervisor: Dr. Sudip K. Garain (IISER Kolkata)	
• Utilised X-ray Data from NuSTAR mission to study stellar mass black holes • Analysed the data Lightcurves, time lag and power density spectra caused by variability • Leading to a connnection between disk-corona-jet of stellar mass black holes	
External Project University of Delhi, Delhi, India	Aug'25 – Present
<i>Stochastic Dynamics of Brownian Motion</i>	
Collaborator: Dr. Raju Roychowdhury (University of Delhi, Delhi)	
• Working on the dynamics of Brownian motion in classical non-conservative systems • Using Green's function to study analytical behavior and correlations • Implementing Monte Carlo simulations for stochastic modeling and statistical analysis	

Summer Internship | *Variable Energy Cyclotron Centre (VECC), Kolkata* *June'24 – July'24*
Development of Carbon and Carbon Monoxide Ion Beams for Si Surface Modification and Nanostructuring with conceptual study on Quantum bit devices
Supervisor: Dr. Prasanta Karmakar (VECC, Kolkata)

- Developed and optimised low-energy ion beams (C^+ , N^+ , N_2^+ , CO^+) using an ECR Ion Source.
- Performed mass separation with dipole magnets and ion implantation on Si(100) substrate.
- Verified through simulated implantation in Si(100) using SRIM.
- Conducted a theoretical study of quantum computing devices and ion beam control systems.

B.Sc. Thesis | *UPES, Dehradun, India* *Aug'23 – May'24*
A Comprehensive Study on Detection Techniques and Trends in Characteristics of Exoplanets
Supervisor: Dr. Rashmi Jain (UPES, Dehradun)

- Used statistical analysis and graphical representation of the archive NASA data on exoplanets and deducing possible trends in exoplanetary properties using TOPCAT.
- Used random forest algorithm to predict a target property using known selected variables using machine learning in Python.
- Lead to a trained model which can fill out gaps in target property cause due to different detection technique using known properties.

TECHNICAL SKILLS

Programming Languages	Python, C++, Fortran, MATLAB, Scilab, LaTeX
Visualization and I/O	HEASARC, Gnuplot, TOPCAT, Aladin, Origin, iSpec
Software Tools	Overleaf, GitHub, SRIM, MS Suite
Operating Systems	Linux, Windows
Online Courses	<i>Astronomy: Exploring Time and Space,</i> <i>From the Big Bang to Dark Energy,</i> <i>Python for Data Science</i>
Training	<i>Summer School on Gravitational-Wave Astronomy</i> ICTS-TIFR, Bengaluru (July 2025)

EXTRACURRICULAR ACTIVITIES

Horizon Science Society (Founding Member) *Jan'23 – May'24*
Head of Marketing and PR

- Organized numerous science society activities among various majors students
- Conducted educational field trip to astronomical observatory of India
- Collaborated with ISRO scientist to organize a workshop on Aditya L1 mission

Sneh Dayal Foundation (Social Internship) *June'22 – July'22*
Group Leader

- Organised health education programs, led research tasks, fundraising and outreach.

COLLABORATORS AND REFEREES

Dr. Arka Chatterjee | *MSc Advisor*

Assistant Professor, Department of Physics

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Dr. Rashi Jain | *BSc Advisor*

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Prof. Santosh Dubey

Professor, Department of Physics

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